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"magnetometer"

Aim: Comparison between magnetic moment by deflection magnetometer

theory:

① **magnet:** a Piece of metal capable of Pulling certain TYPe of materials towards. It has two Poles (S/N)

② **magnetic flux density:** amount of magnetic flux Passing through a unit area that is PerPendicular to The direction of The magnetic field

③ **magnetic force:** is force that magnetic field exerts on a moving electric charge

④ **magnetic moment:** is a measure of The strength and direction of magnet's ability to Produce magnetic field.

Photometer Jolly

Aim → comparison of power illumination between two sources.

law → $\frac{P_1}{P_2} = \frac{r_1^2}{r_2^2}$ ∴ $I = \frac{cP}{r^2}$

Theory :-

1] Photometer Jolley :- A simple device used to measure light intensity by visual comparison.

2] inverse square law → stating that the intensity of any physical effect is invesely proportional to the square of the distance from the source.

3] Power illumination → The amount of light energy emitted or falling on a surface per unit time (lumen)

4] intensity illumination → the amount of luminous flux falling on a unit area of a surface (lux) (lumen)

Tools :-

1- Jolly Photometer
2- Two light source (two candles)
3- Ruler
4- Dark room

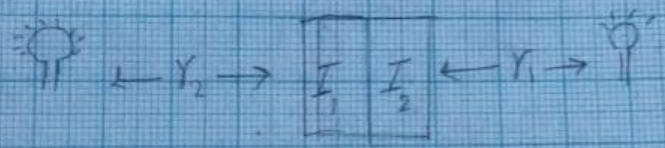
steps :-

1- Place the Jolly Photometer between two light sources
2- Adjust the Photometer's position until both sides sides appear equally bright.

3- Measure the distance of the light sources from the Photometer.

4- calculate the Power ratio using the inverse square law.

law :- $\frac{P_1}{P_2} = \frac{r_1^2}{r_2^2}$



* بناخد r_1 ماشي من البداية لأول منتصف الفوتوميتر ونأخذ نقطة النهاية من r_1 لنحصل على r_2

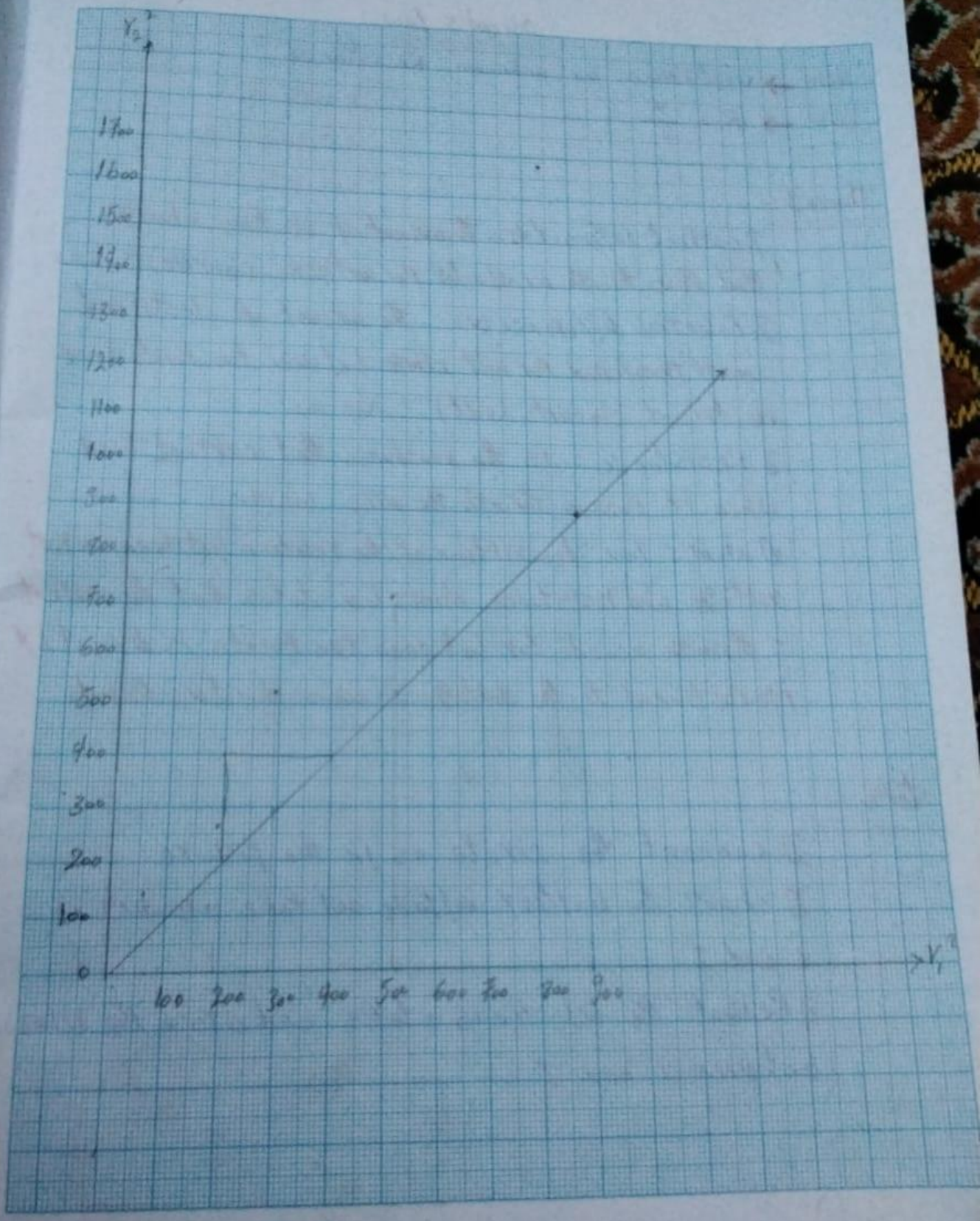
Jolly Photometer

Results:-

r_1 (cm)	r_2 (cm)	r_1^2	r_2^2
8	12	64	144
14	16	196	256
17	23	289	529
23	27	529	729
30	30	900	900
29	41	841	1681

$$\therefore \text{slope} = \frac{\Delta r_1^2}{\Delta r_2^2} = \frac{P_1}{P_2}$$

$$\therefore \frac{P_1}{P_2} = \frac{(400-200)^2}{(400-200)^2} = 1$$


Y_2
1700
1600
1500
1400
1300
1200
1100
1000
900
800
700
600
500
400
300
200
100
0
100
200
300
400
500
600
700
800
900
Y_1

ohm's law

Aim → Determine an unknown Resistance

law → $R = \frac{V}{I}$

Theory :-

1] current intensity (I) :- the amount of electric charge that flows through a conductor per unit time. (ampere) → وحدة

2] Potential difference (V) :- the amount of electrical energy transferred per unit charge between two points in an electrical circuit. (volt) → وحدة القياس

3] Resistance :- is the avoidance that current find when it passes through the wire. (ohm Ω) → وحدة القياس

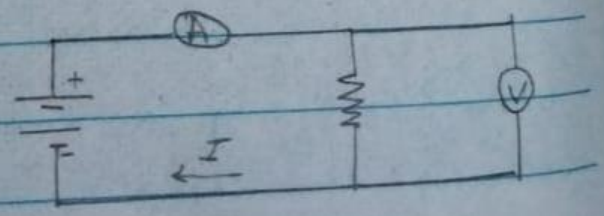
4] ohm's law :- description of the relation between current voltage and resistance. ohm's law states that the current I through a conductor between two points is directly proportional to the voltage V across the two points.

steps :-

1] connect the circle as in the figure

2] chang the battery voltage and take a reading of V and I

3] Repeat the previous steps and draw the relation between V and I



"ohm's law"

Results :-

V	I
0.7	0.04
1.6	0.12
1.8	0.14
1.9	0.16
2.1	0.18
2.4	0.2
3	0.26

$$slope = \frac{1-0.6}{0.1-0.06} = 10\ \Omega = R$$

0.6 الأميتر 0.02

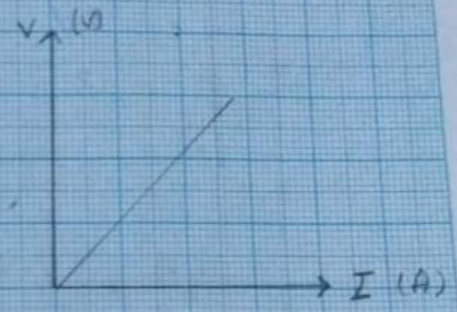
3 الفولتميتر 0.1

صغنن scale

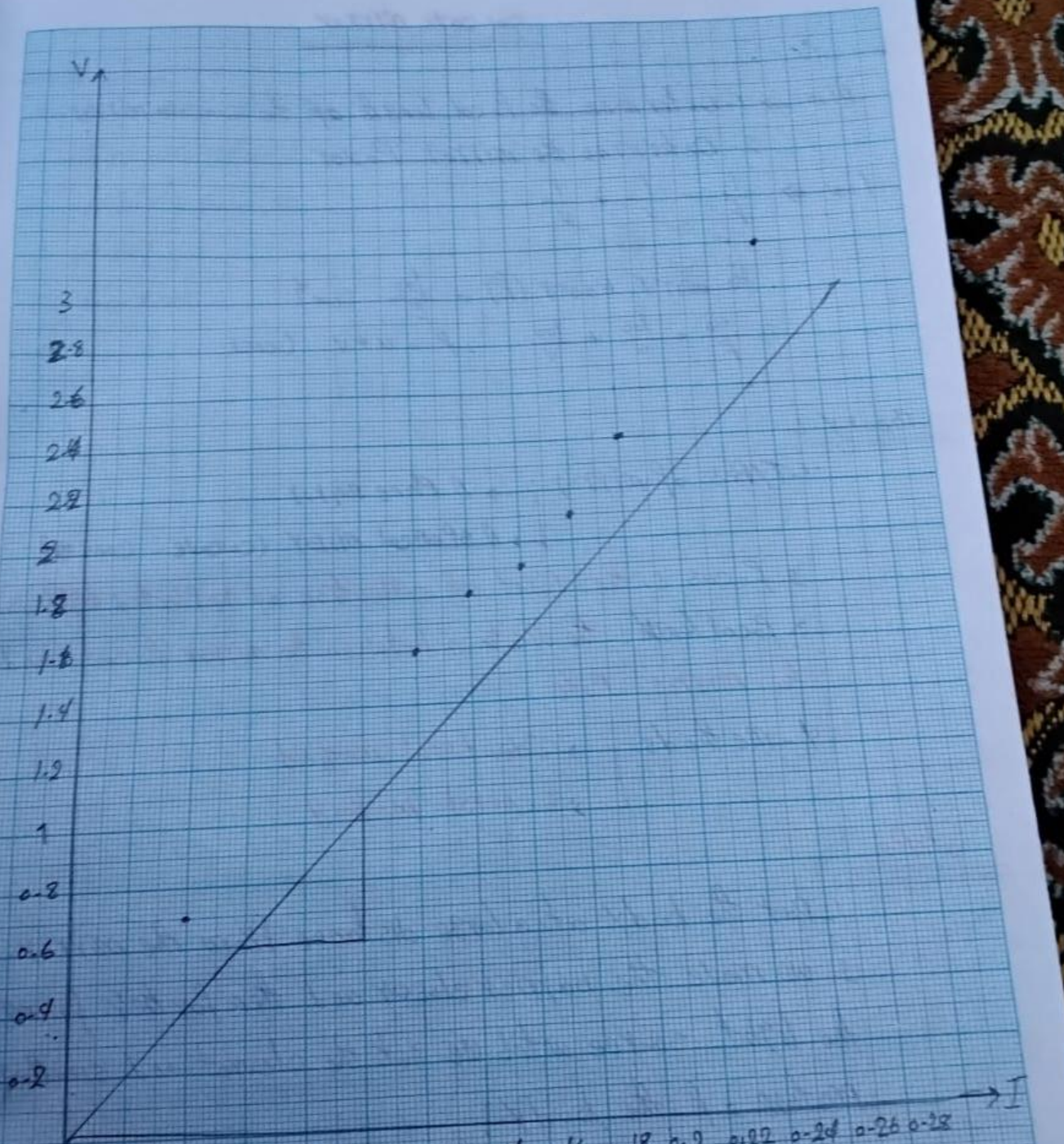
Tools :-

1- Battery
2- Ammeter
3- Voltmeter
4- Connecting wires
5- switch
6- Resistor

law → $R = \frac{V}{I}$



$$\text{slope} = \frac{V}{I} = R \ (\Omega)$$


V
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
I
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.24
0.26
0.28

Concave mirror

Aim → 1- Determine the focal length of the concave mirror.
2- Determine the mirror Power.

law → $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

بعد الجسم عن المرآة ← u
بعد الصاور عن المرآة ← v

- the mirror Power (f) = $\frac{100}{f}$ (cm^{-1})

$\therefore \frac{100}{f} = \frac{100}{u} + \frac{100}{v}$, $f = u + v$ (cm)

Theory :-

1- Types of mirror :
- → A Plane mirror
- → A spherical mirror (concave - convex)

2- focus :- the Point of collection of reflected light rays.

3- focal length :- the distance between the mirror focus and the mirror Pole.

4- Methods
- → Identical method
- → General method

steps

1- Put the body at a large distance from the mirror.

2- we move the barrier closer and then farther from the light source until we get the clearest and smallest picture of the body.

3- the distance between the body and the barrier is the focal length (f)

4- the mirror Power F can be calculated from this relation $F = \frac{100}{f}$

Tools :-

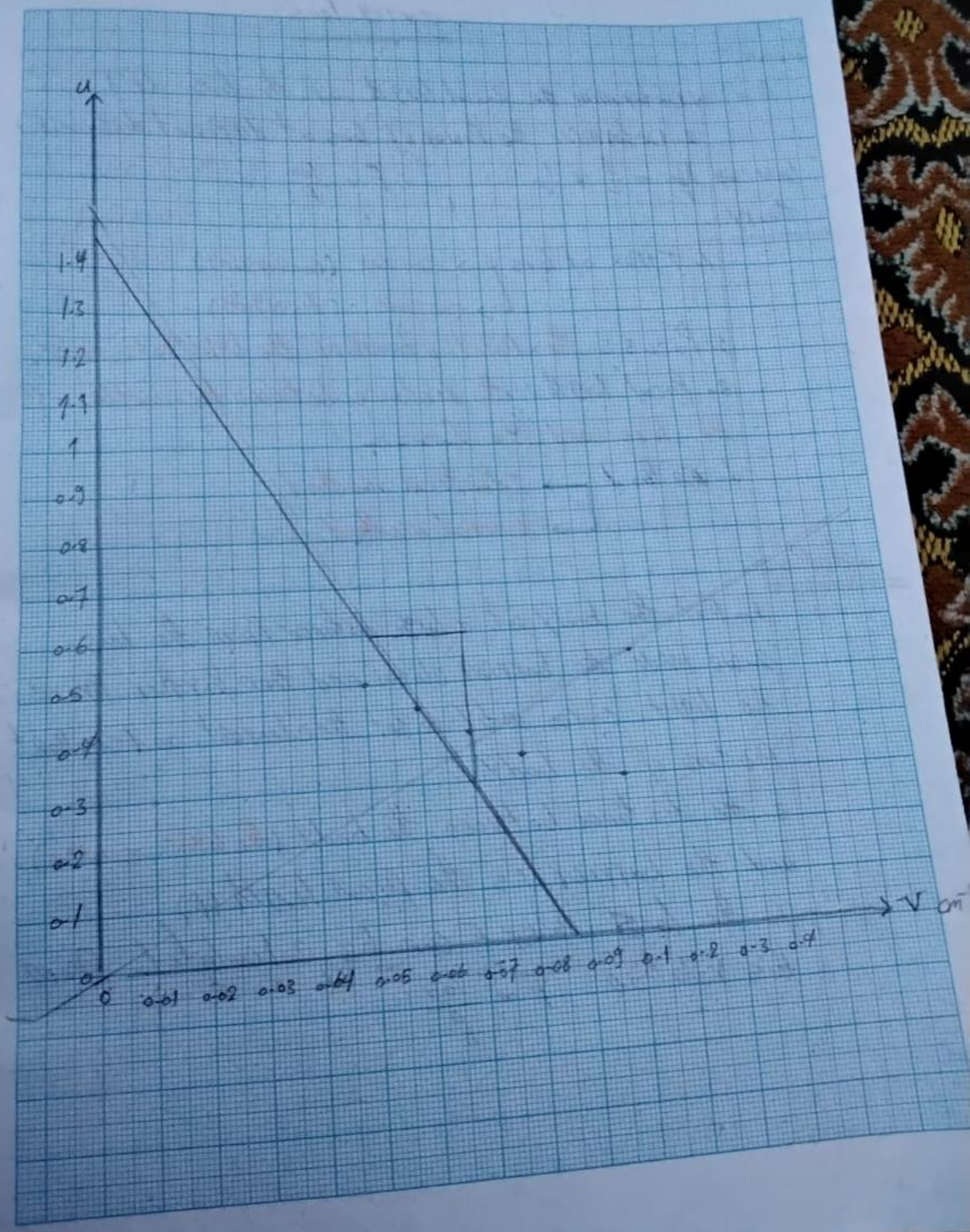
1- concave mirror
2- mirror stand
3- light source
4- ruler
5- barrier

Concave mirror

Results:-

u	v	$\frac{u}{100}$	$\frac{v}{100}$
30	10	0.3	0.1
35	8	0.35	0.08
40	7	0.4	0.07
45	6	0.45	0.06
50	5	0.50	0.05
55	4	0.55	0.04

$$f_{av} = \frac{u+v}{2} = \frac{\quad}{2}$$


u
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1
0.2
0.3
0.4
V

convex lens

Aim → 1- Determine the focal length of the lens (f).
2- Determine the Power of lens at different distance (F).

law ⇒ $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\therefore F = \frac{100}{f}$

theory:-

1- Types of lens → convex (converging).
→ concave (diverging).

2- focus:- the point at which the light rays cross.

3- focal length:- the distance between the center of the lens and its focus.

4- method → Identical method
→ general method

steps:-

1- Put the body at a large distance from the lens.
2- we move the barrier closer and then farther from the light source until we get the clearest and smallest picture of the body.
3- the distance between the body (the light source) and the barrier is the focal length (f).
4- the lens Power F can be calculated from this relation $F = \frac{100}{f}$

Tools :-

1- convex lens
2- light source
3- lens stand
4- barrier
5- ruler

$u \longrightarrow$ المسافة من العدسة لمصدر الضوء

$v \longrightarrow$ المسافة من الحائل للعدسة

* أول طريقة تطابق الصورة على مصدر الضوء ونسميها f_1 ، f_2
وبعدين نزود المسافة ونبدأ نأخذ قراءات u ، v ونرسم علاقة بين u ، v
وبتكون علاقة عكسية بينهم وبعدين نأخذ $f_{av} = \frac{v+u}{2}$

Convex lens

Results :-

u	v	u/100	v/100
45	40	0.45	0.40
50	37	0.50	0.37
55	35	0.55	0.35
60	32	0.60	0.32
65	30	0.65	0.30
70	27	0.70	0.27

v →

Rectangular

Aim → 1- Determine the refractive index of Rectangular "."
2- Achieve the angle of incidence equel to the angle of exit

law ⇒ $n_1 \sin\theta_1 = n_2 \sin\theta_2$ (snell's law) used law $n = \frac{\sin\theta_1}{\sin\theta_2}$

theory :-

1- Reflection :- change in direction of a wave at an interface between 2 different media. "the wave returns into the medium from which it originated."

2- Refraction :- change in direction of a wave passing from medium to another.

3- the optical density :- is a property of a transparent material that measures the speed of the light through the material.

4- Relative refractive index :- It is the ratio of speed of light in one medium to the speed of light in another medium. $\frac{n_1}{n_2} = \frac{v_1}{v_2}$

5- Absolute refractive index :- It is the ratio of light in vacuum to the speed of light in another medium. $n = \frac{c}{v} = \frac{\sin\theta_1}{\sin\theta_2}$

steps :-

1- Put a rectangular on a piece of paper and define its contours

2- Raise the rectangular and draw a line representing an incident ray by angle Φ_1

3- Put a number of Pins on the line.

4- look at the opposite side and you will see the image of the Pins formed

5- Determine the direction of the exit ray by a number of Pins.

6- Raise the rectangular and determine the direction of the refracted ray inside it.

7- measure the angle of incidence, angle of refraction and exit angle.

* Tools:-

1- Rectangular

2- Pins

3- Paper

4- Protractor

"Deflection magnetometer"

Aim ⇒ To compare between the magentic dipole moment of 2 bar magnets.

law ⇒ $\frac{m_1}{m_2} = \frac{\tan\theta_1}{\tan\theta_2}$

theory :-

1- magnetic field :- is a region around a magnetic material or a moving electric charge where magnetic forces can be observed

2- magnetic moment :- is a vector quantity that represents the strength and direction of a magnetic force

Steps

1- Place the magnetic needle on a flat surface

2- set the first magnet at a fixed distance from the needle.

3- measure and record the angle of deflection for first magnet.

4- Repeat the measurement 8 times.

5- Replace the first magnet with the second one keeping the same distance do measurment for the second 8 times.

6- compare the magnetic moments based on the average deflection angles.

Tools :-

1- 2 bar magnets.
2- magnetic needle.
3- compass ruler.

Result :-

Φ_1	Φ_2
N 20° 20°	N 27° 30°
S 25° 25°	S 30° 34°
N 25° 27°	N 30° 30°
S 20 20	S 23° 25°

$$\Phi_{1av} = \frac{20^\circ + 20^\circ + 25^\circ + 25^\circ + 25^\circ + 27^\circ + 20^\circ + 20^\circ}{8} = 22.75^\circ$$

$$\Phi_{2av} = \frac{27^\circ + 30^\circ + 30^\circ + 34^\circ + 30^\circ + 30^\circ + 23^\circ + 25^\circ}{8} = 28.63^\circ$$

$\therefore \tan\Phi_1 = 0.42$ $\qquad$ $\tan\Phi_2 = 0.54$

$$\therefore \frac{m_1}{m_2} = \frac{\tan\Phi_1}{\tan\Phi_2} = \frac{0.42}{0.54} = 0.7$$

Vibration magnetometer

Aim → Comparision between two magnet moment

law ⇒ $\frac{m_1}{m_2} = \frac{T_1^2 + T_2^2}{T_2^2 - T_1^2}$

Theory :-

1- magnetic field :- is a region around a magnetic material or a moving electric charge where magnetic forces can be observed.

2- magnetic moment :- is a vector quantity that represents the strength and direction of a magnetic force.

3- Periodic time :- is the time taken for one complete cycle of a repeating event or motion.

steps :-

1- Mount the magnetometer on a stand and ensure the magnetic needle is suspended by a fine thread so it can move freely in the horizontal plane.

2- Allow the magnetic needle to come to rest pointing toward the magnetic north.

3- Gently displace the needle from its equilibrium and release it to oscillate.

4- use a stop watch to measure the time taken for a certain number of oscillations.

Tools :-

1- stop watch
2- Ruler
3- Bar magnets
4- magneto meter

* باقي الخطوات :-

5- Repeat the timing process several times and take the average to improve accuracy.

6- use the time period to calculate the horizontal component of Earth's magnetic field or the magnetic moment of the magnet using the appropriate formula.

الخطوة دي من مجموعة أووو

Results :-

$N_1 = 3$

$t_1 = 28.06\ s$

$T_1 = \frac{t_1}{N} = \frac{28.06}{3} = 9.35\ s$

$T_1^2 = (9.35)^2 = 87.48\ s^2$

$N_2 = 3$

$t_2 = 47.72$

$T_2 = \frac{t_2}{N} = 15.906\ s$

T_2^2

$T_2^2 = 253.02\ s^2$

$$\therefore \frac{m_1}{m_2} = \frac{T_1^2 + T_2^2}{T_2^2 - T_1^2} = \frac{87.48 + 253.02}{253.02 - 87.48}$$

$$= 2.056$$